

Tailong Xiao

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Research Interests

Quantum Artificial Intelligence (QAI); Machine Learning; Intelligent Quantum Sensing; AI-enhanced Optical Computing and Imaging.

Academic Appointments

Oct. 2023–Present **Assistant Professor**, SJTU, Shanghai, China.

Education

2018-06 **B.E.**, Communication Engineering, Central South University, China.

2023-06 **Ph.D.**, Information and Communication Engineering, SJTU, China.

Selected Publications

1. Xu H, **Xiao T***, Huang J, et al. Toward Heisenberg Limit without Critical Slowing Down via Quantum Reinforcement Learning. *Physical Review Letters*, 2025, 134(12): 120803.
2. **Xiao, T**, Zhai, X., Huang, J. et al. Quantum deep generative prior with programmable quantum circuits. *Communications Physics*, 2024, 7: 276. DOI: 10.1038/s42005-024-01765-9.
3. **Xiao, T**, Zhai, X., Wu, X. et al. Practical advantage of quantum machine learning in ghost imaging. *Communications Physics*, 2023, 6: 171. DOI: 10.1038/s42005-023-01290-1.
4. **Xiao, T**, Fan, J. & Zeng, G. Parameter estimation in quantum sensing based on deep reinforcement learning. *npj Quantum Information*, 8, 2 (2022). DOI: 10.1038/s41534-021-00513-z.
5. **Xiao, T**, Huang, J., Li, H., Fan, J., Zeng, G. Intelligent certification for quantum simulators via machine learning. *npj Quantum Information*, 8, 138 (2022). DOI: 10.1038/s41534-022-00649-6.

* indicates corresponding author. Full list: Google Scholar.

Teaching

- **Engineering Practice and Technological Innovation II (EE0503)**, Spring 2025. Topic: *An Introduction to Quantum Artificial Intelligence* (Undergraduate).

Research Grants & Projects

- **National Key R&D Program of China** (Dec. 2025–Nov. 2030).
- **NSFC Young Scientists Fund** (2025–2027).
- **Lenovo–SJTU Joint Lab Project** (2024–2025).
- **SJTU Young Scholars Project** (2023–2026).

- CCF–Boson Quantum Innovation Fund (2025.03–2025.08).
- National Key Laboratory Open Research Program (2025–2026).

Academic Service

Professional Roles

- Executive Committee Member, CCF Technical Committee on Quantum Computing.

Editorial Work

- Guest Editor, Special Issue “Quantum Computing and Quantum Information Processing”, *Entropy* (MDPI).
Special issue link.
- Editorial Board Member, *Scientific Reports*.

Reviewer

- Physical Review Letters / Applied / Research / A; PRX Quantum.
- New Journal of Physics; Quantum Science and Technology; Journal of Physics A; Machine Learning: Science and Technology; European Journal of Physics.
- npj Quantum Information; Scientific Data; Scientific Reports; Quantum Information Processing.
- Optica Quantum; Optics Express; Optics Letters.

Honors & Awards

- National Scholarship (China), 2017.
- Exemplary Outstanding Student, 2017.
- Hunan Provincial Outstanding Graduate, 2018.
- Yangjiachi Scholarship, 2019.
- CCF Sinan Cup, National Second Prize, 2021.
- First-class Scholarship of State Key Laboratory (Photonics and Communications), 2022.
- CCF Sinan Cup, National First Prize, 2022.
- Outstanding Graduate, 2023.
- CCF Outstanding Doctoral Dissertation in Quantum Computing, 2024.